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ORIGINAL RESEARCH

A novel detection and defense mechanism against false data injection attack in smart grids

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Abstract
As the next generation of green power system, smart grids have gradually enhanced the operation efficiency of power system. Meanwhile, the application of communication and intelligent technologies make the power grid more vulnerable to the emerging cyber-physical attacks, such as the false data injection attack (FDIA). Particularly, the deception property of the FDIA on the output measurement estimation can fool the current security mechanism without triggering an alarm. Motivated by this problem, this paper aims at developing a novel detection and recovery mechanism against FDIA in smart grid. Based on the established state space grid model derived from the three-phase sinusoidal voltage equations, an improved principal component analysis (PCA)-based detection method is proposed. By introducing the mathematical transformation principle method, the detection performance such as detection rate and false positive rate can be improved. To keep the stable running of power system, a genetic optimization algorithm-based linear quadratic regulator (LQR) defense method is developed. In addition, to improve the response performance to external attacks, an artificial intelligence method named genetic optimization algorithm is introduced to optimize the robust performance of the proposed defense method. Finally, the simulation results on the IEEE 6-bus and 118-bus grid system demonstrate the superiority of the proposed genetic algorithm optimization-based LQR defense method.

1 | INTRODUCTION

As the next generation of power system, smart grid integrates the advanced sensing and measurement technology, robust control methods, and high-speed two-way communication network to achieve the goals of reliability, safety, economy, efficiency, and environmental friendliness [1, 2]. Meanwhile, the emerging system characteristics, such as openness and intelligence, bring potential security risks to the safety of smart grid [3]. In particular, the cyber-physical risks caused false data injection attack (FDIA) in recent years have brought great challenges to the current security mechanism of the smart grid. The hackers can cover up the change of system state by injecting false decorated data without triggering an alarm [4]. The recent FDIA has brought great security threats to the stable operation of the power grid, such as Delta Montrose Electric Power Association (DMEA) malicious attack in Colorado, USA, in 2021 and Ransomware attack on Australian power supplier cyber system (CS) energy in 2021. As a result, to confront the emerging enormous challenge caused by FDIA, developing an effective safety mechanism is crucial to keep the stable operation of smart grid system.

As the common network attack, Denial of service attack (Dos) can destroy the power grid system by blocking the transmission of communication channels [5]. In [6], learning-based methods, such as machine and deep learning methods, are presented and analyzed in relation to the detection of cyber attacks in internet of things systems. In [7], a network model that has been subjected to various DoS attacks was proposed to demonstrate cyber security aspect. Noteworthy, the Dos can be detected by using the current safety mechanism. Different from the Dos, the emerging FDIA, such as false data attack, can bypass the current safety mechanism [8, 9]. Therefore, this paper aims at developing a novel detection and defense mechanism against false data attack to ensure the stable running of smart grid systems.

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In recent years, the security research of smart grid has become the top priority of energy departments in various countries. The current security research on FDIA can be mainly divided into two aspects, such as detection-based and defense-based safety mechanism [9]. In the detection-based safety mechanism, scholars have developed different attack characteristics-based attack detection algorithms [10–15]. Considering several cases of severe data-injections with high probabilities of information loss, a Bayesian-based approximated filter (BAF) was proposed to improve the immunity of the monitoring applications against data-injection attacks [10]. In [11], a multisensor temporal prediction scheme was proposed to improve the resilience of wide-area control systems against false data attacks. In [12], Valencia et al. developed a detection and isolation method against Dos by considering the attack impact on the communication channels. In [13], the attack feature of Dos on the power system is analyzed. Based on the invasion pathway of the FDIA, Cao et al. proposed a focal-loss-lightgbm ensemble classifier to detect the injected FDIA in cyber-physical system [14]. Taking the feature of GPS spoofing attack into account, a detection method is proposed by using the changes of estimated line reactance [15]. In [16], a deep-learning-based detection and classification mechanism is proposed by considering the impact on the communication networks caused by the cyber attacks. By using the extensive knowledge of the physical grid model, an online detection framework against FDIA is proposed to protect the automatic generation control [17]. To sum up, the above detection-based detection methods can effectively protect the safety of power system. However, one can find that the effectiveness of the above methods depends on the assumption that the attack sequence is known. For this reason, some scholars further have proposed defense-based safety mechanism to protect the power system [18–24]. In [18], an interval state estimation-based defense method against FDIA is proposed by considering the lower- and upper bounds of state variables. To ensure the safety of power system, a novel resilient defense framework is proposed based on the Luenberger observer including an artificial neural network [19]. By analyzing the impact of FDIA on the load frequency, a novel defense control strategy is developed [20]. In [21], a robust hybrid defense framework including multiple sub-controllers is established. To enhance the security of the existing defense method, a resilient control method is proposed to control the load frequency control of interconnected power systems [22]. Considering time delays and data-packet losses, a dynamic event triggering defense mechanism against FDIA is developed in [23]. To make sure the state of the systems stable, a sliding mode observer-based defense method is proposed to control the generator load frequency [24]. Generally, the selection of an LQR feedback matrix depends on the experience of designer. For this reason, it is difficult to ensure that the designed LQR control performance is optimal. In a sum, there still exist some problems for the above defense methods to be solved as follows: (1) How to solve the limitation of attack detector on defense methods; (2) How to address the limitation of parameter matrix selection for the robust controller.

Motivated by the above problems, this paper develops a genetic optimization algorithm-based LQR defense method against FDIA in smart grid. Firstly, a state space model derived from the three-phase sinusoidal voltage equations is established. Considering the influence of FDIA on the internal voltage, an improved principal component analysis (PCA)-based detection method is proposed. Then, the improved LQR is designed to respond to the injected malicious attack quickly. In additional, a genetic optimization algorithm is introduced to address the limitation of the precomputed design parameter matrix. Based on the global search ability of genetic algorithm, the optimal parameter weight matrix can be obtained. The main contributions of this work are given as follows:

1. An improved PCA-based detection method is developed to solve the influence of attack detector on defense methods. Compared with the work in [25], the proposed method can enhance the detection rate and reduce the false-positive rate by introducing the mathematical transformation principle method.
2. A genetic optimization algorithm-based defense method is proposed to address the limitation of the precomputed design parameter matrix. Compared with the existing defense works in [26] and [27], the developed method can quickly eliminate the influence of FDIA on voltage.

The outline of this work is organized in the following. Section 2 gives the state-space model derived from the three-phase sinusoidal voltage equations and the description of FDIA. In Section 3, the genetic optimization algorithm-based LQR defense framework is proposed. The simulation experiments are provided to show the effectiveness of the proposed algorithm in Section 5. Finally, Section 6 presents the conclusion and discussion.

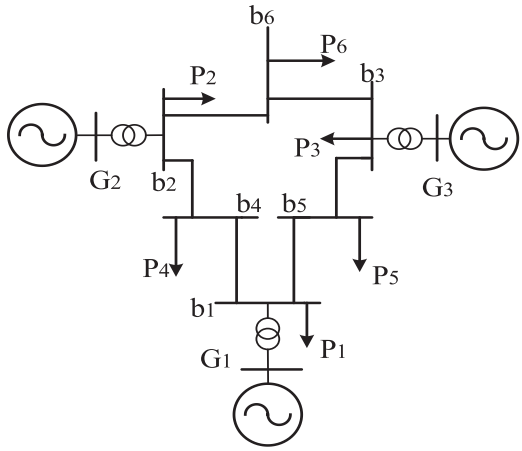

FIGURE 1 IEEE 3-generator and 6-bus grid system.

2 | THEORY

2.1 | Grid state model

As shown in Figure 1, the 3-generator and 6-bus power system is described. As usual, the smart sensors can obtain the corresponding the measured voltage, current, and phase. Therefore,

we can obtain the state-space model of one voltage measurement derived from the three-phase voltage formula as follows [28].

$$V(k) = A_v * \cos(wk)\cos(\phi) + A_v * \sin(wk)\sin(\phi) \quad (1)$$

where A_v denotes the amplitude of the voltage, ωk denotes the angular frequency, and ϕ denotes the phase.

It is assumed that the angular frequency ωk is relatively constant over time, the state variables such as amplitude and phase are considered. Then, the corresponding the grid model on three-phase voltage is obtained as follows:

$$x_k = Ax_{k-1} + Bu_k + w_k \quad (2)$$

$$y_k = Hx_k \quad (3)$$

where $x_k = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^T$, $x_1 = A_v \times \cos\phi$, and $x_2 = A_v \times \sin\phi$ are state variable at epoch k, A_v represents the amplitude of the voltage, $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $w(k)$ is the process noise, which follows Gaussian distribution, $H = (\cos\omega k - \sin\omega k)$ is the measurement matrix.

2.2 | Description of false data injection attack

According to the work in [29], the traditional detection criterion of bad data is described as follows:

$$\begin{cases} \xi_k = \|\mathbf{z}_k - Hx_k\| < \eta & \text{Normal} \\ \xi_k = \|\mathbf{z}_k - Hx_k\| \geq \eta & \text{Abnormal} \end{cases} \quad (4)$$

where ξ_k is the measurement residual, $\mathbf{z}_k$ is the measurement output, η the prior threshold, which is chosen such that the error rate is less than 5%. To minimize the false positive caused by the noise, we set the threshold to 3σ in [30].

To cheat the above detection method, hackers can design a bank of malicious attack sequence y_k^a meeting the following condition as

$$\begin{aligned} \xi_k^a &= \left\| z_k^a - H\hat{x}_k^a \right\| \\ &= \left\| (z_k + y_k^a) - (H\mathbf{x}_k + c) \right\| \\ &= \left\| (z_k - H\hat{x}_k) + (y_k^a - Hc) \right\| \\ &\leq \left\| z_k - H\hat{x}_k \right\| + \left\| y_k^a - Hc \right\| \end{aligned} \quad (5)$$

where ξ_k^a, $\mathbf{z}_k^a$, and $\mathbf{x}_k^a$ are the measurement residual, measurement output, and state estimation under FDIA, respectively, $\mathbf{c}$ is the increment of state caused by the attack.

If $y_k^a = Hc$, one can find that Equation (5) indicates the $\xi_k^a = 0$. In other words, if the attacker can inject the above attack sequence y_k^a, the common detection method based on $\chi 2$-detector can be bypassed. It is assumed that the attacker can inject the malicious attack sequence in Equation (5) at $k = 50$ s on the IEEE 3-generator and 6-bus grid system, the correspond running state and detection results can be obtained, as shown in Figure 2. It is noted that the above parameters are taken from the work in [28].

As shown in Figure 2a, one can find that the voltage amplitude has been changed by injecting the malicious attack sequence $k = 50$ s. However, the detection result in Figure 2b indicates that the detection method based on the $\chi 2$-detector can be fooled. For this reason, how to ensure the normal running of power system under FDIA is an urgent problem.

2.3 | Problem formulation

In a sum, the goal of this paper aims at developing a novel detection and recovery mechanism to ensure the stable operation of grid system, which is described as follows:

$$\begin{cases} x_k = Ax_{k-1} + Bu_k + w_k + Ff_k \\ y_k = Hx_k \end{cases} \quad (6)$$

where f_k denotes the attack vector in Equation (5), F is the attack selected matrix.

In the next section, the proposed security mechanism including the improved PCA detection method and the genetic optimization algorithm-based LQR defense method is developed to address the security problem in smart grid.

3 | THE DETECTION AND DEFENSE MECHANISM AGAINST FALSE DATA INJECTION ATTACK IN SMART GRID

In this section, a detection and defense mechanism against FDIA is proposed. Based on the constructed grid model, an improved PCA-based detection method is proposed. Then, an improved LQR controller is constructed to eliminate fluctuations caused by FDIA. To address the limitation of the precomputed design parameter matrix, a genetic optimization algorithm is introduced based on the global search ability.

3.1 | Improved PCA-based detection method

Under normal running condition, we can collect a set of sensor history data matrix on the voltage of generators as $X_{n\times m}$ (n is the number of voltage sampling, m is the number of sensors).

Then, we can standardize the data $X_{n\times m}$ and obtain the sample covariance as

$$\Upsilon = \frac{1}{n} X^T X \quad (7)$$

From Equation (7), the singularly decouple Υ as

$$\Upsilon = \Lambda\Gamma\Lambda^T = \begin{bmatrix} \hat{\Lambda} & \tilde{\Lambda} \end{bmatrix} \begin{bmatrix} \hat{\Gamma} & 0 \\ 0 & \tilde{\Gamma} \end{bmatrix} \begin{bmatrix} \hat{\Lambda} & \tilde{\Lambda} \end{bmatrix}^T \quad (8)$$

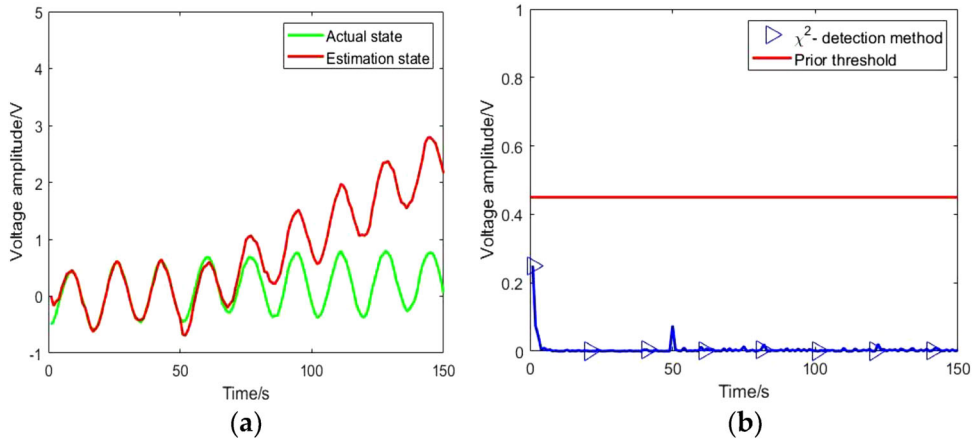

FIGURE 2 The simulation results under detection method based on $\chi 2$-detector. (a) The state change under malicious attack. (b) The detection results under the detection method based on $\chi 2$-detector.

where Λ is the diagonal matrix, where diagonal elements satisfy $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m$, $\Gamma_{m\times m}$ denotes the orthogonal matrix, U is the a columns of $\Gamma_{m\times m}$ containing the primary meta information, and $\bar{U}$ is the $m-a$ columns of $\Gamma_{m\times m}$ containing the primary meta information.

Based on Equations (7) and (8), we can decompose a generic sample $x(k)$ as follows:

$$x = \hat{x} + \tilde{x} \tag{9}$$

where $\hat{x} = \hat{\Lambda}\hat{\Lambda}^T$ denotes the score matrix, $\tilde{x} = \tilde{\Lambda}\tilde{\Lambda}^T x$ is the residual subspace.

According to the work in [31], the two detection indexes Square Prediction Error δ_{spe} and the Hotelling's δ_T are given as follows:

$$\delta_{spe} = x^T \tilde{\Lambda}\tilde{\Lambda}^T x \leq \delta(\alpha) \tag{10}$$

$$\delta_T = \hat{\Lambda}\hat{\Gamma}^{-1}\hat{\Lambda}^T \leq \tau(\alpha) \tag{11}$$

where $\delta(\alpha)$ and $\tau(\alpha)$ are the precomputed threshold, which are derived assuming a specified confidence level $(1-\alpha) \times 100$.

Based on Equations (10) and (11), a combined detection index is developed as follows:

$$\delta_{\text{new}} = \frac{\delta_{\text{spe}}}{\delta(\alpha)} + \frac{\delta_{\text{T}}}{\tau(\alpha)} \tag{12}$$

Before dealing with the historical data, a cognitive geometric transformation is introduced to enhance the detection performance. Of note, the detailed mathematical transformation principle can be seen in [32].

Based on the improved PCA, the pseudo code of the proposed detection method against FDIAs is presented in Algorithm 1.

ALGORITHM 1 Improved PCA-based detection algorithm against false data injection attack in smart grid

1. SDM→Sensor data matrix,
2. VOG→Voltage of generators,
3. CGTD→Cognitive geometric transformation of data,
4. SC→Sample covariance,
5. DT→Detection threshold,
6. SDM ←Obtain(VOG),
7. CGTD←transform (SDM),
8. SC←(CGTD),
9. δ_{new}←extract(SC),
10. For $\kappa = 1 \tau o N$
11. if δ_{new} < DT then, attack
12. elseif no attack.
13. **eigenvalues**: abnormal data ←extract(δ_{new}).

3.2 | Genetic optimization algorithm-based LQR defense method

3.2.1 | Design of the improved LQR controller

Before designing the improved LQR controller, an assumption is given as follows:

Assumption 1. [33] For the linear system in Equation (6), (A, B) are controllable and $\text{rank}(B, F) = rank(B)$.

It is noted that Assumption 1 indicates that there exists $\text{Im}(F) = \text{Im}(B)$, which is equal to the existence of B such that

$$(I - BB^*)F = 0 \tag{13}$$

To ensure the stable running of power system under FDIA, we design the following improved LQR controller as

$$u_k = Ky_k - BF\hat{f}_k \tag{14}$$

Compared with the traditional LQR controller, the designed $BF\hat{f}_k$ is introduced to eliminate the influence of FDIA.

From Equations (6) and (13), one can obtain

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k + Ff_k + w_k \\ &= (A + BKH)x_k - BB^* \hat{f}_k + Ff_k + w_k \\ &= A'x_k + \begin{bmatrix} I & F \end{bmatrix} \begin{bmatrix} w_k \\ e'_k \end{bmatrix} \\ &= A'x_k + F'\bar{w}_k \end{aligned} \tag{15}$$

where $A' = A + BKH$, $e'_k = \hat{f}_k - f_k$, $F' = [I \quad F]$ and $\bar{w}_k = \begin{bmatrix} w_k \\ e'_k \end{bmatrix}$.

To ensure the robust performance of the improved LQR controller, a theorem is given as follows:

Theorem 1. *Assuming there exist symmetric matrices $P = P^T$, $J = J^T$, $N = \begin{bmatrix} N_{11} & 0 \\ N_{21} & N_{22} \end{bmatrix}$, and M satisfying the following LMI:*

$$\begin{bmatrix} -\bar{P} & 0 & \bar{A}N + \bar{B}\begin{bmatrix} M & 0 \end{bmatrix} & \bar{F}' \\ * & -I & \bar{H}N & F' \\ * & * & \bar{J} - N - N^T & 0 \\ * & * & * & -\iota I \end{bmatrix} \tag{16}$$

where $\bar{A} = T_H^{-1}AT_H$, $\bar{B} = T_H^{-1}B$, $\bar{F}' = T_H^{-1}F'$, $\bar{H} = HT_H$, $P = T_H^{-1}\bar{P}T_H$, $J = T_H^{-1}\bar{J}T_H$, then, the system in Equation (15) under FDIA with the following robust performance index:

$$\sum_{k=0}^{\infty} \|y_k\|^2 < \iota \sum_{k=0}^{\infty} \|\bar{w}_k\|^2 \tag{17}$$

where the controller gain is given as follows:

$$K = MN_{11}^{-1} \tag{18}$$

Proof of Theorem 1. Since $P = T_H\bar{P}T_H^T$ and $J = T_H\bar{J}T_H^T$, one can obtain

$$\bar{P} = T_H^{-1}PT_H^{-T} \text{and} \bar{J} = T_H^{-1}JT_H^{-T} \tag{19}$$

Then, Equation (16) can be rewritten as

$$\begin{bmatrix} -T_H^{-1}PT_H^{-T} & 0 & T_H^{-1}AT_HN + T_H^{-1}B\begin{bmatrix} M & 0 \end{bmatrix} & T_H^{-1}F' \\ * & -I & HT_HN & F' \\ * & * & T_H^{-1}JT_H^{-T} - N - N^T & 0 \\ * & * & * & -\iota I \end{bmatrix} \tag{20}$$

By using the matrix

$$W = \begin{bmatrix} T_H & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & T_H & 0 \\ 0 & 0 & 0 & I \end{bmatrix},$$

we can obtain

$$\begin{bmatrix} -P & 0 & AT_HNT_H^T + B\begin{bmatrix} M & 0 \end{bmatrix} & T_H^TF' \\ * & -I & HT_HNT_H^T & F' \\ * & * & J - T_HNT_H^T - T_HN^TT_H^T & 0 \\ * & * & * & -\iota I \end{bmatrix} \tag{21}$$

By using Equation (17) and

$$N = \begin{bmatrix} N_{11} & 0 \\ N_{21} & N_{22} \end{bmatrix},$$

Equation (15) can be rewritten as

$$\begin{bmatrix} -P & 0 & AT_HNT_H^T + BKT_HNT_H^T & T_H^TF' \\ * & -I & HT_HNT_H^T & F' \\ * & * & J - T_HNT_H^T - T_HN^TT_H^T & 0 \\ * & * & * & -\iota I \end{bmatrix} \tag{22}$$

Based on the fact that $J - T_HNT_H^T - T_HN^TT_H^T \geq -T_HNT_H^TJ^{-1}T_H^TN^TT_H$ and the following matrix and its transpose, we can obtain

$$\tilde{W} = \begin{bmatrix} I & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & (T_HNT_H^T)^{-1} & 0 \\ 0 & 0 & 0 & I \end{bmatrix},$$

$$\begin{bmatrix} -P & 0 & A + BKH & F' \\ * & -I & H & F' \\ * & * & -J^{-1} & 0 \\ * & * & * & -\iota I \end{bmatrix} \tag{23}$$

Then, pre-multiplying and post-multiplying of Equation (17) with $[x_k^T \ \bar{w}_k^T]$ and its transpose, we can obtain

$$x_{k+1}^TPx_{k+1} - x_k^TJx_k + y_k^Ty_k - \iota\bar{w}_k^T\bar{w}_k < 0 \tag{24}$$

Equation (24) indicates that the proof of **Theorem 1** is complete. □

Obviously, **Theorem 1** ensures the stability of the proposed LQR controller. Namely, the proposed LQR controller can

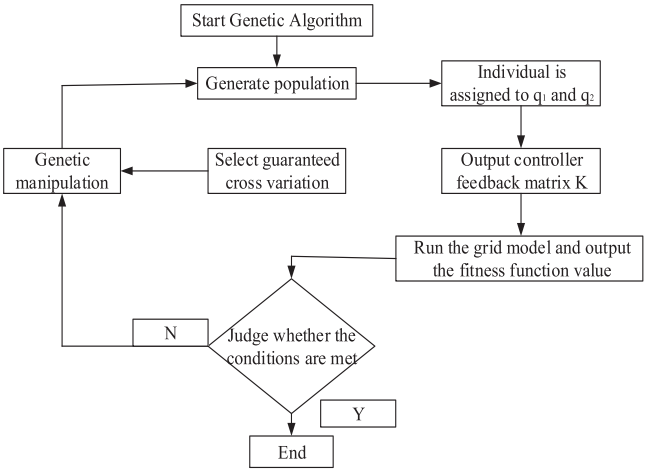

FIGURE 3 Process of optimizing LQG controller by genetic algorithm.

eliminate the effect of FDIA to keep the power system stable running. In addition, to minimize the robust performance of the LQR controller, **Theorem 1** is equal to the following optimization problem:

$$\min_{P,J,M} \iota \tag{25}$$

3.2.2 | Genetic optimization algorithm-based LQR defense method

Generally, the selection of matrices P,J, and M is calculated artificially in advance. For this reason, the precomputed design parameter matrix limits the robust performance of the improved LQR controller. In this section, a genetic optimization algorithm is introduced based on the global search ability to address this problem in the following.

To realize the optimal solution of weight matrix P,J, and M, the process of optimizing linear quadratic regulator (LQG) controller by genetic algorithm is given, as shown in Figure 3.

Significantly, the optimization objective of fitness function value L of genetic algorithm is set as follows:

$$\begin{aligned} &\min imize\ L = \frac{RMS[x_1].S}{RMS_{pas}[x_1]} + \frac{RMS[x_2].S}{RMS_{pas}[x_2]}, \\ &S = (q_1, q_2),\ st \begin{cases} \dfrac{RMS[x_1]}{RMS_{pas}[x_1]} < 1 \\ \dfrac{RMS[x_2]}{RMS_{pas}[x_2]} < 1 \end{cases} \end{aligned} \tag{26}$$

where RMS denotes the root mean square value of data, RMS_{pas} denotes the optimization performance of amplitude of the voltage and the phase, and S is the weighting coefficient.

Then, the pseudo code of the proposed defense method against FDIAs is presented in Algorithm 2.

ALGORITHM 2 Genetic optimization-based LQR controller defense algorithm against false data injection attack in power system

1. PSM→Power system model,
2. LQRC→LQR controllers,
3. GA→Genetic algorithm,
4. CFMS→Controller feedback matrix set,
5. MCFMS→Min controller feedback matrix set,
6. FFV→Fitness function value,
7. LQRC ←Design(PSM),
8. CFMS ←Design(LQRC),
9. FFV←Design(GA),
10. For All parameter matrices P, J, M.
11. If FFV is min then MCFMS ←Design(CFMS),
    Else
    Repeat Step 7- Step 9;
    end if
8. end for
9. **eigenvalues**: MCFMS ←extract(CFMS).

4 | RESULTS

In this section, simulation results on IEEE 3-generators and 6-bus grid system are given to demonstrate the effectiveness of the developed detection and defense mechanism against FDIA. The related simulation parameters are set as follows: A 60-Hz, 0.5-V sinusoidal voltage signal with random Gaussian noise is generated, other parameters are taken from the work in [28].

4.1 | Case 1: improved PCA-based detection of voltage under false data injection attack

Assume that a hacker can inject a bank of false data injection to modify the sensor data of generator 1 at $t = 20$ s. By using the proposed detection Algorithm 1, the detection results under δ_{spe} and δ_T are obtained, as shown in Figure 4.

Obviously, one can find that although the detection index δ_{spe} can detect attacks in Figure 4a, detection index δ_T cannot detect the false data timely in Figure 4b. For this reason, it is difficult to judge whether there are data exceptions based on the above detection indexes. By using the combined detection index δ_{new}, the corresponding detection results and contribution rate are obtained, as shown in Figure 5.

As shown in Figure 5a, one can find that the injected false data can be identified timely. In addition, we can obtain the contribution rate that causes data exceptions, as shown in Figure 5b. Obviously, the injected false data are generated by the sensor data of generator 1. As shown in Table 1, the detection rate under different methods is given. Compared with the work in [25], one can find that the improved PCA-based detection method can enhance the detection performance against FDIA. And the comparative analyses of missed detection rate

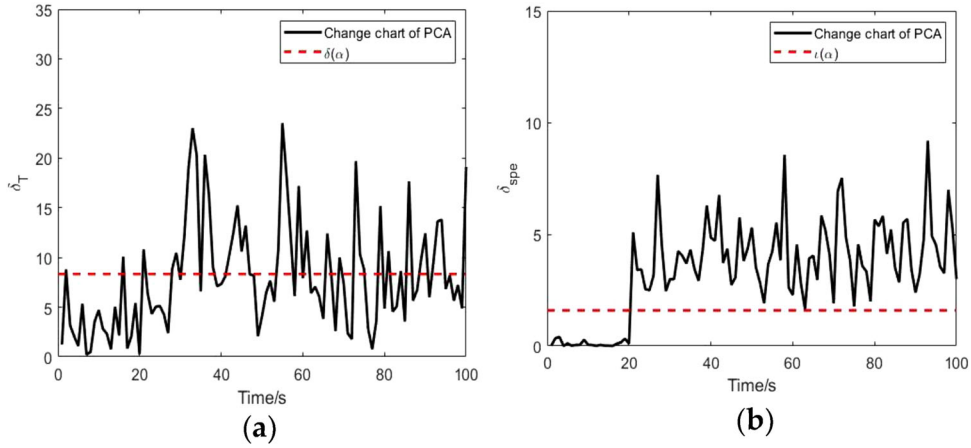

FIGURE 4 The detection results of voltage response under detection indexes δ_{spe} and δ_T.

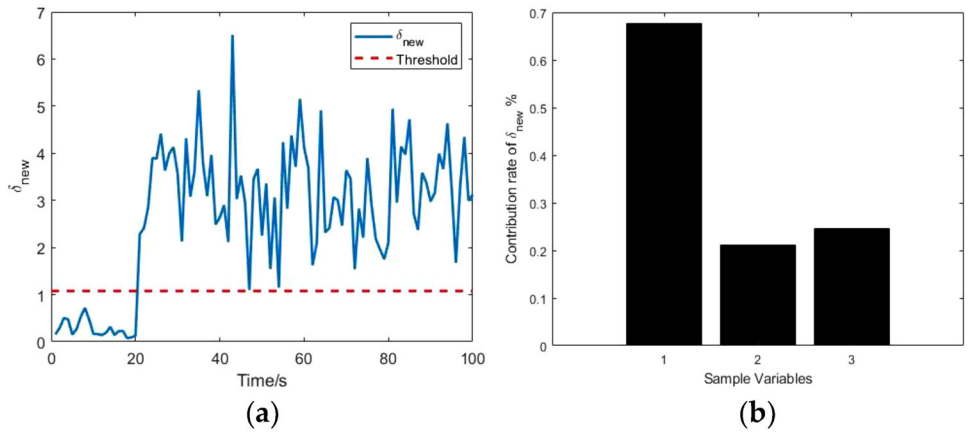

FIGURE 5 The detection results of voltage response under detection index δ_{new} and corresponding contribution rate.

TABLE 1 Comparison of detection performance

Sample	Detection rate under [25]	detection rate under Algorithm 1
20	83.1%	91.7%
120	89.5%	98.2%

TABLE 2 Comparison of detection performance

Sample	Missed detection rate [25]	missed detection rate by Algorithm 1
20	15.8%	8.6%
120	13.9%	5.5%

under different methods are obtained, as shown in Table 2. Obviously, the proposed detection Algorithm 1 can reduce the missed detection rate. Furthermore, the simulation results in Figure 6 demonstrate the probability of missed detection rate versus attack intensity.

4.2 | Case 2: control of voltage without false data injection attack

Assuming that there does not exist the FDIA in power system, the common LQR controller and genetic optimization-based LQR controller are designed to control the voltage by using the proposed Algorithm 1. The corresponding the global genetic generation of fitness value and voltage response are obtained, as shown in Figures 7 and 8.

Obviously, one can find that the proposed genetic algorithm converges globally as the number of iterations increases, as shown in Figure 7. Namely, when the number of iterations is 6, the optimal solution of the LQR feedback matrix can obtain the optimal solution. As shown in Figure 8, one can find that the peak value of voltage response under LQR+GA has been improved. Namely, the robust performance under LQR+GA is better than that under LQR. As shown in Table 3, the performance index of voltage response can be enhanced 27.49%. In other words, the proposed genetic optimization-based LQR controller can quickly respond to the change of voltage by comparing with the common LQR controller.

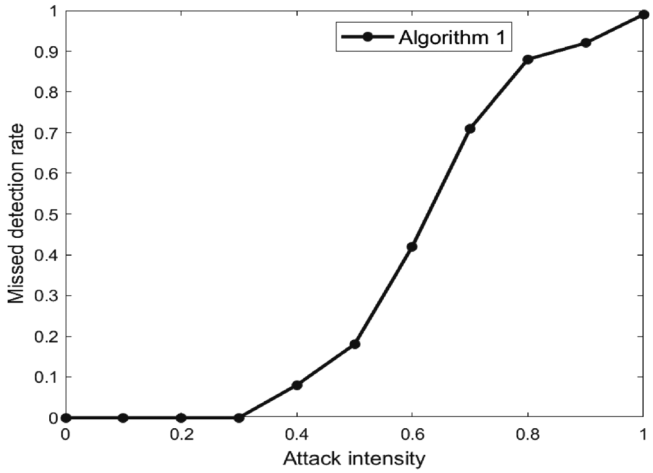

FIGURE 6 The probability of missed detection rate versus attack intensity.

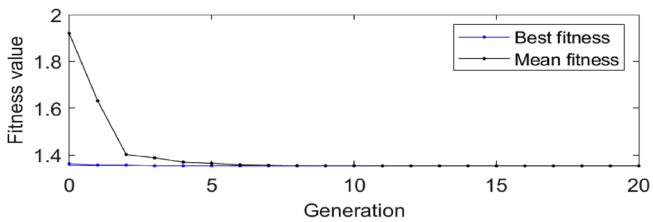

FIGURE 7 The global genetic generation for fitness value.

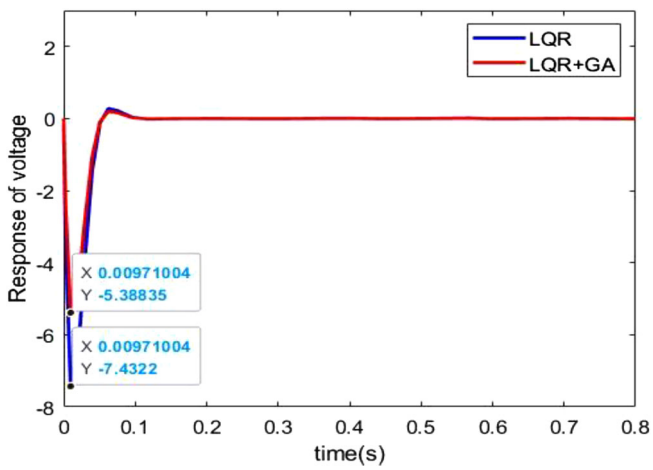

FIGURE 8 The simulation results of voltage response under LQR and LQR+GA. GA, genetic algorithm.

4.3 | Case 3: control of voltage with false data injection attack

In this section, we consider the effectiveness of the proposed genetic optimization-based LQR controller against FDIA. It is assumed that hackers can inject the malicious attack into the generator 1 at $t = 20$ s. By using the proposed Algorithm 2, a bank of LQR and LQR+GA controller are designed to ensure the stable running of voltage of power system. The corresponding voltage response under FDIA is obtained, as shown in Figures 9 and 10.

TABLE 3 Comparison of performance index by using the root mean square value

Performance index	LQR	LQR+GA	Improvement
Voltage response (V)	−7.4322	−5.38835	27.49%

GA, genetic algorithm

TABLE 4 Analysis of detection performance on the IEEE 118-bus grid system under Algorithm 1

Accuracy rate	missed detection rate	train and test time(s)	optimization time of LQR(s)
97.85%	5.6%	39	36

The simulation results in Figure 9 demonstrate the voltage responses on the first, second, and third generators under FDIA. Obviously, one can find that the voltage of the second and third generators does not get attacked by hackers. However, the voltage of the first generator is abnormal, as shown in Figures 9a and 9b. Namely, the hacker can successfully attack the voltage of generator 1 and destroy the stable running of power system, as shown in Figure 9c. Furthermore, the deviation of voltage in Figure 10a indicates that the voltage of the first generator is abnormal. However, there does not exist the deviation voltage on other second and third generators.

The simulation results in Figure 11 demonstrate the control effect of voltage under different defense methods. As shown in Figure 11a, the LQR controller can respond to the change of voltage. However, the effect caused by FDIA cannot be eliminated by the LQR controller. Meanwhile, the simulation results in Figure 11b show that the proposed genetic optimization-based LQR controller defense method can quickly respond to the change of voltage caused by FDIA. Meanwhile the effect caused by FDIA can be eliminated. In other words, the proposed genetic optimization-based LQR controller defense method can effectively and quickly protect the grid system.

4.4 | Case 4: performance analysis on the IEEE 118-bus grid system

To demonstrate the detection and defense performance, simulations on the IEEE 118-bus are given as follows. Then, we evaluate the defense performance by the time required for the system to recovery. As shown in Table 4, one can find that the detection and missed detection rate are 97.85% and 5.6%, respectively. Therefore, the proposed Algorithm 1 can effectively detect the injected false data. Meanwhile, one can find that the train and test time is 39 s, optimization time of LQR is 36 s. As shown in Table 5, one can find that the recovery time of the proposed genetic optimization-based LQR controller defense

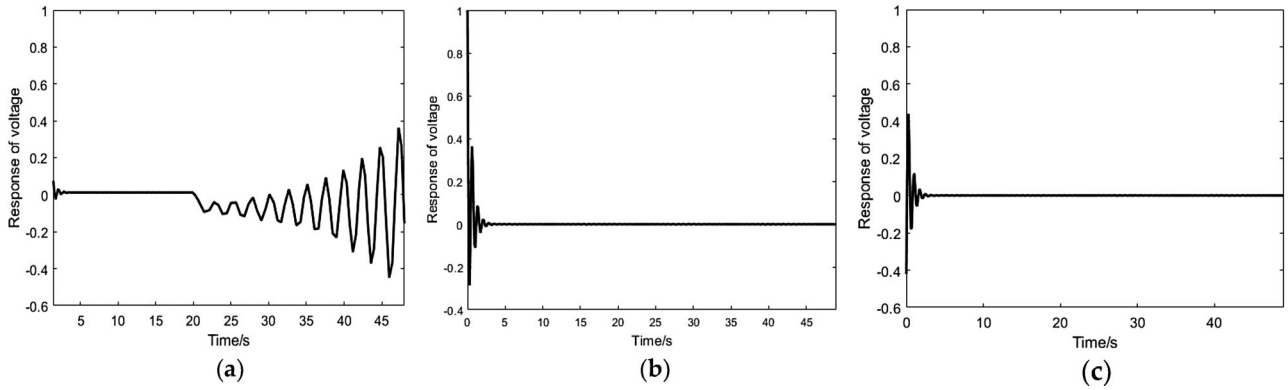

FIGURE 9 The change of voltage response under false data injection attack without defense method. (a) First generator, (b) second generator, (c) third generator.

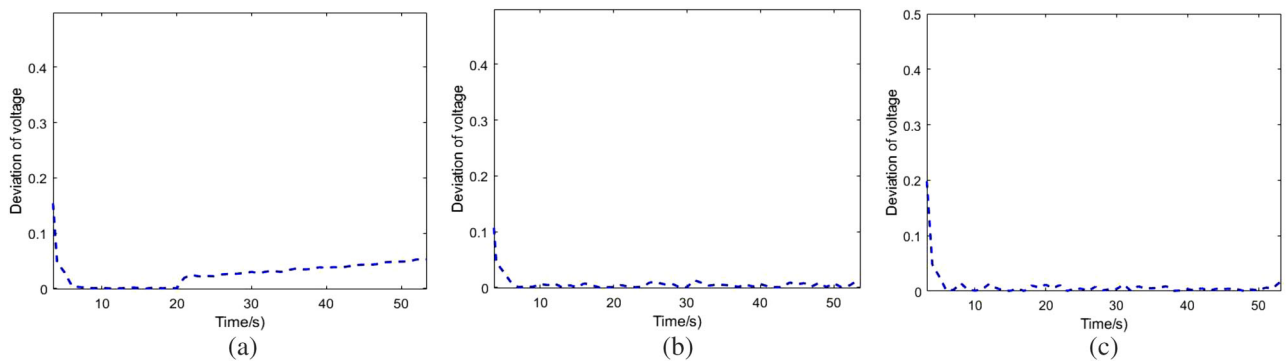

FIGURE 10 The deviation of voltage under false data injection attack without defense method. (a) First generator, (b) second generator, (c) third generator.

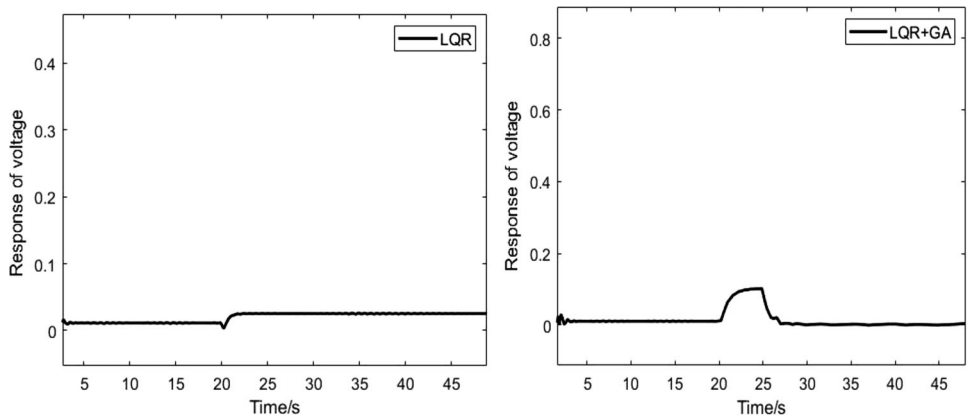

FIGURE 11 The change of voltage response under false data injection attack with two defense methods and. (a) LQR controller-based defense method, (b) proposed genetic optimization-based LQR controller defense method.

method is 7.16 s. Obviously, compared with other defense works in [26] and [27], the proposed defense method can eliminate the impact of the malicious attack on the voltage in the shortest time to ensure the safe operation of the power grid. It is worth noting that the recovery time in the literature has been given in [34].

5 | CONCLUSIONS AND DISCUSSION

This paper proposes a novel detection and defense mechanism against FDIA in smart grid. Through the constructed physical dynamics model, an improved PCA-based detection method is proposed. By using the detection results, a genetic optimization

TABLE 5 Comparison of recovery time under false data injection attack

Method	Required recovery time(s)
Method in [26]	16.52
Method in [27]	7.55
Proposed method	7.16

algorithm-based LQR controller defense method is designed to compensate the change of FDIA on the voltage. Based on the global search ability of genetic algorithm, a genetic optimization algorithm is introduced to address the limitation of the precomputed design parameter matrix. Finally, the simulation results on the IEEE 6-bus and 118-bus grid system verify the performance of the proposed detection and recovery method.

To improve the defense performance against FDIA, the following problems needed to consider. (1) How to shorten the defense response time to attacks through detection. (2) How to deal with the influence of model uncertainty on the controller.

AUTHOR CONTRIBUTIONS

Conceptualization, J.C. and B.G.; methodology, B.G; data curation, J.C; writing - original draft preparation, B.G; writing - review and editing, J.C. and B.G; visualization, B.G.; All authors have read and agreed to the published version of the manuscript.

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INSTITUTIONAL REVIEW BOARD STATEMENT

Not applicable.

INFORMED CONSENT STATEMENT

Not applicable.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Not applicable.

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